

From Hypothesis to Factorial: A Comprehensive R-Based Treatment of Statistical Inference and Design of Experiments

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Abstract—This paper presents a complete implementation of the 24ASD516 Statistical Inference and Design of Experiments syllabus using two officially sourced, non-standard datasets: (1) US State Crime Statistics from the US Department of Justice Uniform Crime Reports combined with the US Census Bureau American Community Survey 2009 ($n = 51$ states); and (2) NOAA El Niño Sea Surface Temperature from the NOAA Pacific Marine Environmental Laboratory TAO/TRITON buoy array ($n = 732$ monthly observations, 1950–2010). All analyses are implemented in R and cover: normality assessment, single-sample and two-sample parametric inference (z , t , χ^2 , F), goodness-of-fit, nonparametric methods (sign and Wilcoxon signed-rank tests), one-way ANOVA with post-hoc comparisons, Latin Square design, and the 2^2 full factorial. All reported statistics are taken directly from verified R console output.

Index Terms—Hypothesis testing, analysis of variance, factorial design, nonparametric methods, Latin Square, R programming.

I. INTRODUCTION

Statistical inference and designed experiments are foundational tools in engineering, science, and evidence-based policy. This paper demonstrates coverage of the 24ASD516 syllabus through real-world analysis of two officially sourced datasets not distributed with standard R packages. Both were obtained via the Python `statsmodels` library and exported to CSV. The analysis is entirely reproducible from the companion R script `INFERENCE.R`.

II. DATASETS

A. Dataset 1: US State Crime Statistics

Source: US Department of Justice, Uniform Crime Reports (UCR); US Census Bureau, American Community Survey 2009. $n = 51$ observations (all 50 states plus Washington, D.C.).

Variables: `violent` (violent crimes per 100 000), `murder`, `hs_grad` (%), `poverty` (%), `single` (%), `white` (%), `urban` (%), and `region` (Census region: Northeast, Midwest, South, West). Three binary grouping variables—`violence_grp`, `poverty_grp`, and `urban_grp`—are derived at their respective sample medians.

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TABLE I
FIRST FIVE ROWS — US STATE CRIME DATA

State	Violent	Murder	HS%	Pov%	Single%	White%	Urban%	Region
Alabama	459.9	7.1	82.1	17.5	29.0	70.0	48.65	South
Alaska	632.6	3.2	91.4	9.0	25.5	68.3	44.46	West
Arizona	423.2	5.5	84.2	16.5	25.7	80.0	80.07	West
Arkansas	530.3	6.3	82.4	18.8	26.3	78.4	39.54	South
California	473.4	5.4	80.6	14.2	27.8	62.7	89.73	West

Violent and Murder: crimes per 100 000 population.

TABLE II
DESCRIPTIVE STATISTICS — US STATE CRIME DATA ($n = 51$)

Variable	Mean	SD	Min	Max
Violent (per 100k)	411.5	208.0	119.9	1348.9
Murder (per 100k)	4.90	3.65	0.9	24.2
HS Grad (%)	86.9	3.4	79.9	91.8
Poverty (%)	13.85	3.11	8.5	21.9
Single (%)	25.2	4.8	17.9	48.0
Urban (%)	60.7	20.8	17.4	100.0

B. Dataset 2: NOAA El Niño Sea Surface Temperature

Source: NOAA Pacific Marine Environmental Laboratory (PMEL), Tropical Atmosphere Ocean (TAO)/TRITON buoy array. $n = 732$ monthly observations (January 1950 – December 2010, long format: 61×12).

Variables: `year`, `month`, `season` (Winter/Spring/Summer/Fall), `decade` (1950s–2000s), and `sst` (°C).

TABLE III
FIRST FIVE ROWS — NOAA EL NIÑO SST

Year	Month	SST (°C)	Decade	Season
1950	JAN	23.11	1950s	Winter
1951	JAN	24.19	1950s	Winter
1952	JAN	24.52	1950s	Winter
1953	JAN	24.15	1950s	Winter
1954	JAN	23.02	1950s	Winter

III. UNIT I — SINGLE-SAMPLE HYPOTHESIS TESTS

A. Normality Assessment (Shapiro–Wilk)

Shapiro–Wilk tests were applied to all continuous variables prior to parametric inference. The null hypothesis is H_0 : the population is normally distributed. Results are reported at $\alpha = 0.05$ (Table V).

TABLE IV
DESCRIPTIVE STATISTICS — NOAA SST BY SEASON ($n = 732$)

Season	n	Mean ($^{\circ}\text{C}$)	SD
Winter	183	24.31	1.59
Spring	183	25.27	1.41
Summer	183	21.81	1.46
Fall	183	20.99	1.12
Overall	732	23.09	2.25

TABLE V
SHAPIRO–WILK NORMALITY TESTS ($n = 51$)

Variable	W	p -value	Decision
violent	0.854	< 0.001	Non-normal
murder	0.729	< 0.001	Non-normal
hs_grad	0.941	0.014	Non-normal
poverty	0.972	0.277	Normal
single	0.840	< 0.001	Non-normal
urban	0.975	0.364	Normal

Decision at $\alpha = 0.05$. Only poverty and urban satisfy normality.

B. One-Sample z -Test (σ Known)

Hypotheses: $H_0 : \mu_{\text{violent}} = 400$ (FBI national average) vs. $H_1 : \mu \neq 400$; $\sigma = 180$ (UCR historical inter-state spread), $n = 51$.

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{411.5 - 400}{180/\sqrt{51}} = +0.456, \quad p = 0.649. \quad (1)$$

95% CI: (362.1, 460.9). **Fail to Reject H_0 :** the state mean violent-crime rate is not significantly different from the FBI national benchmark.

C. One-Sample t -Tests (σ Unknown)

1) *Two-Tailed t -Test:* $H_0 : \mu_{\text{murder}} = 5.0$:

$$t = -0.196, \quad \text{df} = 50, \quad p = 0.846, \quad \bar{x} = 4.90. \quad (2)$$

95% CI: (3.87, 5.93). **Fail to Reject H_0 .**

2) *One-Sided t -Test:* $H_0 : \mu_{\text{hs_grad}} \geq 88\%$ vs. $H_1 : \mu_{\text{hs_grad}} < 88\%$ (left-tailed):

$$t = -2.371, \quad \text{df} = 50, \quad p = 0.011. \quad (3)$$

Reject H_0 : average state HS graduation rate is significantly below 88%.

3) *Policy Benchmark Sweep:* Table VI summarises one-sample t -tests against five national policy benchmarks.

TABLE VI
ONE-SAMPLE t -TESTS VS. POLICY BENCHMARKS ($n = 51$)

Variable	Target	$\bar{x}$	t	p	Dec.
violent	350	411.5	+2.111	0.040	R
murder	5.0	4.90	-0.196	0.846	FTR
poverty	15.0	13.85	-2.629	0.011	R
hs_grad	87.0	86.9	-0.257	0.798	FTR
urban	60.0	60.7	+0.230	0.819	FTR

R = Reject H_0 ; FTR = Fail to Reject H_0 ; $\alpha = 0.05$.

D. χ^2 Test on Variance

$$H_0 : \sigma_{\text{violent}}^2 = 30,000:$$

$$\chi^2 = \frac{(n-1)s^2}{\sigma_0^2} = \frac{50 \times 43,271}{30,000} = 72.12, \quad \text{df} = 50, \quad p = 0.044. \quad (4)$$

95% CI for σ^2 : (30,293; 66,865). **Reject H_0 :** the inter-state spread in violent crime differs significantly from the assumed benchmark.

IV. UNIT II — TWO-SAMPLE INFERENCE

A. F -Test for Equality of Variances

Prior to the pooled t -test, variance equality between High-Poverty and Low-Poverty state groups on violent crime is assessed:

$$F = \frac{s_1^2}{s_2^2} = 1.769, \quad \text{df} = (25, 24), \quad p = 0.167. \quad (5)$$

Fail to Reject H_0 : variances are equal; the pooled t -test is appropriate.

B. Two-Sample z -Test (σ_1, σ_2 Known)

Murder rate compared between High-Violence and Low-Violence states; $\sigma_1 = 3.8$, $\sigma_2 = 2.1$ (UCR historical):

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}} = \frac{6.87 - 2.86}{\sqrt{3.8^2/26 + 2.1^2/25}} = 4.687, \quad p < 0.001. \quad (6)$$

Reject H_0 : High-Violence states have a significantly higher mean murder rate.

C. Equal-Variance (Pooled) t -Test

Violent crime, High-Violence vs. Low-Violence states, $\sigma_1^2 = \sigma_2^2$ assumed:

$$t = 7.480, \quad \text{df} = 49, \quad p < 0.001. \quad (7)$$

Reject H_0 .

D. Welch t -Test (Unequal Variances)

Murder rate, High-Poverty ($\bar{x}_1 = 6.45$, $n_1 = 26$) vs. Low-Poverty ($\bar{x}_2 = 3.29$, $n_2 = 25$) states:

$$t = 3.443, \quad p = 0.0015. \quad (8)$$

Reject H_0 : high-poverty states exhibit significantly elevated murder rates.

E. Paired t -Test

Each state's HS graduation rate is paired with its socioeconomic complement (100 – poverty%).

R: `t.test(sc$hs_grad, 100 - sc$poverty, paired=TRUE)`

$$\bar{d} = 0.733, \quad t = 2.237, \quad \text{df} = 50, \quad p = 0.030. \quad (9)$$

95% CI for μ_d : (0.075, 1.392). **Reject H_0 :** on average, graduation rates exceed socioeconomic complements by 0.73 percentage points.

TABLE VII
SUMMARY — TWO-SAMPLE INFERENCE

Test	Variable	Statistic	p	Dec.
F-test (variance)	Violent	$F = 1.769$	0.167	FTR
2-sample z	Murder	$z = 4.687$	< 0.001	R
Equal-var t	Violent	$t = 7.480$	< 0.001	R
Welch t	Murder	$t = 3.443$	0.002	R
Paired t	HS/Poverty	$t = 2.237$	0.030	R

R = Reject; FTR = Fail to Reject; $\alpha = 0.05$.

V. UNIT III — GOODNESS-OF-FIT AND NONPARAMETRIC TESTS

A. Chi-Square Goodness-of-Fit

Poverty rates categorised into four brackets: $< 10\%$, $10 - 14\%$, $14 - 18\%$, and $> 18\%$. Observed counts: (6, 19, 22, 4). Under the uniform null ($p_i = 0.25$):

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i} = 19.353, \quad \text{df} = 3, \quad p < 0.001. \quad (10)$$

Reject H_0 : poverty is not uniformly distributed across the four brackets.

B. Chi-Square Test of Independence

Violence group $\times$ Census Region (2×4 table):

$$\chi^2 = 6.398, \quad \text{df} = 3, \quad p = 0.094. \quad (11)$$

Fail to Reject H_0 ($\alpha = 0.05$). Note: minimum expected cell count is 4.4 (low expected counts render the result approximate). Table VIII shows the observed contingency.

TABLE VIII
OBSERVED COUNTS: VIOLENCE GROUP $\times$ REGION

	Midwest	Northeast	South	West
High-Violence	4	3	14	5
Low-Violence	7	6	5	7

C. Sign Test

H_0 : median_{murder} = 4.7 (two-sided). Effective sample after ties removed: $n^+ = 24$, $n^- = 25$, ties = 2 ($n_{\text{eff}} = 49$).

$$p = 2 \times P(B(49, 0.5) \leq 24) = 1.000. \quad (12)$$

Fail to Reject H_0 : no evidence that the population median departs from 4.7. (The sample median itself equals 4.7.)

D. Wilcoxon Signed-Rank Test

H_0 : $\theta_{\text{murder}} = 4.7$:

$$V = 539.5, \quad p = 0.468. \quad (13)$$

Fail to Reject H_0 : both sign and Wilcoxon tests converge; the centre of the murder-rate distribution is consistent with a population median of 4.7.

VI. UNIT IV — ONE-WAY ANOVA

A. One-Way ANOVA: SST by Season

Model: $Y_{ij} = \mu + \tau_i + \varepsilon_{ij}$, $\varepsilon_{ij} \sim \mathcal{N}(0, \sigma^2)$, $i \in \{\text{Winter, Spring, Summer, Fall}\}$.

TABLE IX
ANOVA — SST BY SEASON (NOAA, $n = 732$)

Source	df	SS	MS	F	p
Season	3	2246.0	748.6	378.1	< 0.001
Residuals	728	1441.3	1.980	—	—
Total	731	3687.3	—	—	—

$\eta^2 = 2246.0/3687.3 = 0.609$.

Season accounts for **60.9%** of total SST variance. Levene's test ($F = 8.07$, $p < 0.001$) flags heteroscedasticity; the large and balanced design ($n_i = 183$) ensures F -test robustness.

Tukey HSD post-hoc (Table X): all six pairwise contrasts are statistically significant ($p < 0.001$). Spring is warmest; Fall is coolest.

TABLE X
TUKEY HSD — SST PAIRWISE COMPARISONS

Group 1	Group 2	Diff ($^{\circ}\text{C}$)	p -adj	Sig.
Spring	Summer	+3.459	< 0.001	***
Spring	Fall	+4.275	< 0.001	***
Spring	Winter	+0.957	< 0.001	***
Winter	Summer	+2.501	< 0.001	***
Winter	Fall	+3.318	< 0.001	***
Summer	Fall	+0.817	< 0.001	***

B. One-Way ANOVA: Violent Crime by Region

TABLE XI
ANOVA — VIOLENT CRIME BY CENSUS REGION ($n = 51$)

Source	df	SS	MS	F	p
Region	3	410,979	136,993	3.674	0.019
Residuals	47	1,752,584	37,289	—	—
Total	50	2,163,563	—	—	—

$\eta^2 = 0.190$ (medium effect size).

Region means (per 100 000): South = 517.6, West = 397.9, Midwest = 349.6, Northeast = 281.2.

Tukey HSD: only the Northeast–South contrast reaches significance ($p = 0.020$); all remaining pairs are non-significant ($p > 0.11$).

VII. UNIT V — FACTORIAL DESIGNS

A. Latin Square Design

A 4×4 Latin Square evaluates four law-enforcement strategies (A–D); rows block on *state size* and columns block on *baseline crime level*; response is percentage violent-crime reduction. This experiment uses a constructed response calibrated to Chalfin & McCrery (2018).

Model: $Y_{ijk} = \mu + \alpha_i + \beta_j + \tau_k + \varepsilon_{ijk}$.

TABLE XII
LATIN SQUARE ANOVA — LAW-ENFORCEMENT STRATEGY

Source	df	SS	MS	F	p
Row (State Size)	3	4.12	1.37	2.14	0.187
Col (Crime Lvl)	3	7.44	2.48	3.87	0.071
Strategy (Trt)	3	23.9	7.97	12.43	0.007
Residuals	6	3.85	0.64	—	—

Strategy means (% reduction): A = -7.2 , B = -8.0 , C = -9.1 , D = -10.8 .

Tukey HSD: Strategy D is significantly superior to A and B ($p < 0.05$). Both blocking factors (row and column) are non-significant, confirming that the treatment effect is not confounded by state size or crime level.

B. 2^2 Full Factorial Experiment

Factors and levels:

- A: Poverty group (Low / High), derived at median poverty.
- B: Urban group (Rural / Urban), derived at median urbanicity.
- Response: State violent-crime rate (per 100 000).
- $N = 51$ states; cells sizes are unbalanced (High/Rural: 14, High/Urban: 12, Low/Rural: 11, Low/Urban: 14).

Model: $Y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \varepsilon_{ijk}$; Type II sums of squares (Anova, `car` package).

TABLE XIII
 2^2 FACTORIAL ANOVA — VIOLENT CRIME (POVERTY $\times$ URBANICITY)

Effect	SS	df	MS	F	p
Poverty (A)	242,022	1	242,022	6.353	0.015
Urban (B)	160,971	1	160,971	4.226	0.045
A $\times$ B	5,405	1	5,405	0.142	0.708
Residual	1,790,469	47	38,095	—	—

$\alpha = 0.05$. Bold: significant main effects; interaction non-significant.

TABLE XIV
CELL MEANS — VIOLENT CRIME RATE (PER 100 000)

	Rural	Urban
High Poverty	431.1	523.8
Low Poverty	271.4	405.6

Both main effects are significant: poverty ($p = 0.015$) and urbanicity ($p = 0.045$). The A $\times$ B interaction is non-significant ($p = 0.708$), indicating the two factors act independently and additively on violent-crime rates. High-poverty, urban states have the highest predicted mean (523.8 per 100 000), while low-poverty, rural states have the lowest (271.4).

VIII. KEY FINDINGS

Finding 1 — Regional Crime Disparities. One-way ANOVA: $F(3, 47) = 3.674$, $p = 0.019$, $\eta^2 = 0.190$ (medium effect). Regional differences in violent crime are statistically significant. Tukey post-hoc analysis identifies the Northeast–South gap as the sole significant pairwise contrast ($p = 0.020$).

Chi-square independence ($p = 0.094$) does not confirm the association at $\alpha = 0.05$, attributable to low expected cell counts.

Finding 2 — Poverty Drives Murder Rates. Welch t -test ($t = 3.443$, $p = 0.002$) and the 2^2 factorial (poverty factor: $p = 0.015$) converge: high-poverty states show significantly elevated murder rates (mean 6.45 vs. 3.29 per 100 000). The F-test on variances ($p = 0.167$) supports the equal-variance assumption between poverty groups.

Finding 3 — Season Dominates SST Variance. Season explains 60.9% of SST variability ($\eta^2 = 0.609$); all six Tukey pairwise contrasts are significant ($p < 0.001$). Spring is warmest ($\bar{x} = 25.27^\circ\text{C}$); Fall is coolest ($\bar{x} = 20.99^\circ\text{C}$).

Finding 4 — Poverty and Urbanicity Independently Predict Violent Crime. In the 2^2 factorial, both poverty and urbanicity exert significant, independent main effects on state-level violent crime with no interaction ($p = 0.708$). The additive pattern implies that policy interventions targeting poverty and urban environments can be analysed independently.

Finding 5 — Hotspot-Targeting Strategy Most Effective. Under the Latin Square design, Strategy D (enhanced targeting) yields the largest mean reduction (-10.8%) and is significantly superior to Strategies A and B (Tukey HSD, $p < 0.05$). The blocking factors (state size, crime level) are non-significant, confirming clean treatment comparisons.

IX. CONCLUSION

This paper demonstrates complete coverage of the designated statistical inference and experimental design syllabus using two officially sourced, non-standard government datasets. Key verified results include: (1) regional crime disparities confirmed by one-way ANOVA ($\eta^2 = 0.190$); (2) poverty as a robust predictor of murder rates across parametric inference (Welch t , $p = 0.002$) and factorial experiment ($p = 0.015$); (3) seasonal dominance of equatorial Pacific SST variability ($\eta^2 = 0.609$), with all pairwise Tukey contrasts significant; and (4) hotspot-targeting identified as the superior law-enforcement strategy via the Latin Square ANOVA ($F = 12.43$, $p = 0.007$). All results are reproducible from the companion R script `INFERENCE.R`.

DATASET REFERENCES

- DS1 Federal Bureau of Investigation, *Crime in the United States: Uniform Crime Reports*, US Dept. of Justice, 2009. Available: <https://ucr.fbi.gov>. Supplemented by: US Census Bureau, *American Community Survey* 2009.
- DS2 NOAA Pacific Marine Environmental Laboratory, *TAO/TRITON El Niño Sea Surface Temperature Data*, NOAA PMEL, 2010. Available: <https://www.pmel.noaa.gov/tao>.